MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, GERMANIUM, HIGH-FREQUENCY

TYPE 2N384

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a PNP, germanium, high-frequency transistor.

1.2 Physical dimensions. See figure 1 (TO-44).

1.3 Maximum ratings.

P_T 1/ T_A = 25° C	V_{CBO}	V_{EBO}	V_{CEO}	I_C	T_{stg}
mW	Vdc	Vdc	Vdc	mAdc	°C
120	-40	-0.5	-20	-10	-65 to +100

1/ Derate linearly 1.6 mW/° C for T_A >25° C.

1.4 Primary electrical characteristics.

Limits	C_{obo} V_{CB} = -12 Vdc I_E = 0 100 kHz≤f≤1 MHz	h_{fe} V_{CE} = -12 Vdc I_E = 1.5 mAdc	NF V_{CB} = -12 Vdc I_E = 1.5 mAdc f = 50 MHz
	pf		db
Min	---	20	---
Max	3	175	10

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

FSC 5961

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STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The transistors shall be of the design, construction, and physical dimensions specified in figure 1.

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III.

3.5 Marking. The following marking specified in MIL-S-19500 may be omitted from the body of the transistor at the option of the manufacturer:

(a) Country of origin.
(b) Manufacturer's identification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500 and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of groups A, B, and C inspections.

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

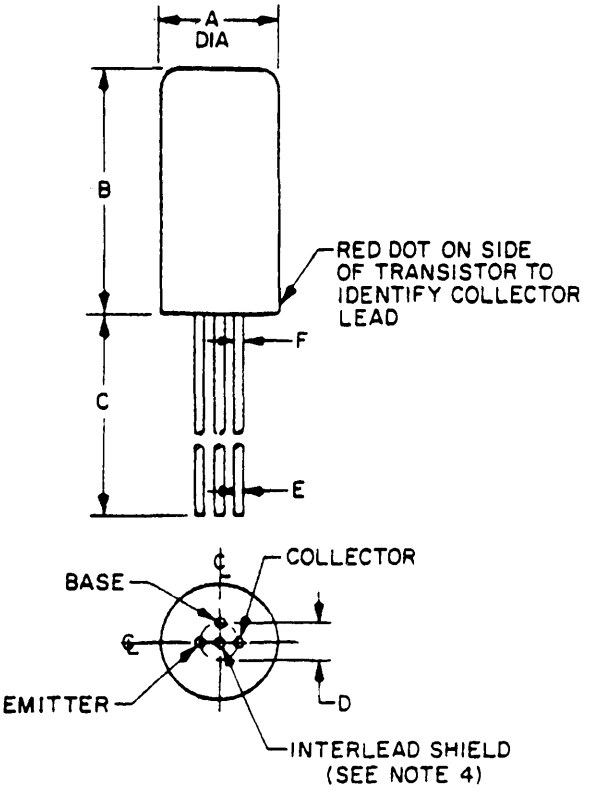
4.3.3 Group C inspection. Group C inspection shall consist of the examinations and tests specified in table III. This inspection shall be conducted on the initial lot and thereafter every 6 months during production.

4.3.4 Group B and group C life-test samples. Samples that have been subjected to group B, 340-hours life-test, may be continued on test to 1,000 hours in order to satisfy group C life-test requirements. These samples shall be predesignated, and shall remain subjected to the group C 1,000-hour acceptance evaluation after they have passed the group B, 340-hour acceptance criteria. The cumulative total of failures found during 340-hour test and during the subsequent interval up to 1,000 hours shall be computed for 1,000 hour acceptance criteria.

4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I, II, and III.

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Subgroup 2, see TABLE III herein; steps 1, 2, and 3.



LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A	---	.240	---	6.10	
B	---	.405	---	10.29	
C	1.500	1.750	38.10	44.45	5
D	.064	.080	1.63	2.03	
E	.016	.021	.41	.53	2,5
F	.016	.019	.41	.48	3,5

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. Measured in the zone beyond .250 (6.35 mm) from the seating plane.
3. Measured in the zone .050 (1.27 mm) and .250 (6.35 mm) from the seating plane.
4. This lead is electrically connected to the case.
5. All four leads.

FIGURE 1. Physical dimensions of transistor type 2N384 (TO-44).

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4.4.1 Collector saturation current. The specified reverse dc voltage shall be applied between the collector and base terminals. The emitter shall be dc open-circuited. The collector saturation current shall then be measured.

TABLE I. Group A inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			10				
Visual and mechanical examination	2071			---	---	---	---
Subgroup 2			5				
Breakdown voltage, collector to base	3001	Bias cond. D; I_C = -50 μAdc		BV_{CBO}	-40	---	Vdc
Collector to emitter cutoff current	3041	Bias cond. D; V_{CE} = -20 Vdc		I_{CEO}	---	-1.0	mAdc
Breakdown voltage, emitter to base	3026	Bias cond. D; I_E = -12 μAdc		BV_{EBO}	0.5	---	Vdc
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -12 Vdc		I_{CBO}	---	-12	μAdc
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = -12 Vdc; I_E = 1.5 mAdc		h_{fe}	20	175	---
Collector saturation current	---	V_{CB} = 1.5 Vdc (see 4.4.1)		I_{CB}(sat)	---	-8.0	μAdc
Subgroup 3			10				
Noise figure	3246	V_{CB} = -12 Vdc; I_E = 1.5 mAdc; f = 50 MHz; R_g = 30 ohms; R_L = 2,000 ohms		NF	---	10	db
Open-circuit output capacitance	3236	V_{CB} = -12 Vdc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz		C_{obo}	---	3	pf
Magnitude of small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = -12 Vdc; I_E = 1.5 mAdc; f = 10 MHz		$\|h_{fe}\|$	2.5	8	---
Small-signal power gain	3256	V_{CB} = -12 Vdc; I_E = 1.5 mAdc; f = 50 MHz; R_g = 30 ohms; R_L = 2,000 ohms		G_{pe}	13	---	db
Small-signal power gain	3256	V_{CB} = -4 Vdc; I_E = 1.5 mAdc; f = 50 MHz; R_g = 30 ohms; R_L = 2,000 ohms		G_{pe}	11	---	db

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TABLE II. Group B inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			20				
Physical dimensions	2066	(see figure 1)		---	---	---	---
Subgroup 2			15				
Solderability	2026			---	---	---	---
Thermal shock (temperature cycling)	1051	Test cond. B except T_{max} = 95° C and time at temperature extremes = 15 minutes min.		---	---	---	---
Thermal shock (glass strain)	1056	Test cond. A except T_{max} = 95° C.		---	---	---	---
				---	---		
Moisture resistance	1021	Omit initial conditioning		---	---	---	---
End points:							
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -40 Vdc		I_{CBO}	---	-50	μAdc
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = -12 Vdc; I_E = 1.5 mAdc		h_{fe}	20	175	---
Subgroup 3			15				
Shock	2016	Nonoperating: 500 G; 1.0 msec, 5 blows in each orientation: X_1, Y_1, Y_2, and Z_1		---	---	---	---
Vibration, variable frequency	2056			---	---	---	---
Constant acceleration	2006	5,000 G, in each orientation: X_1, Y_1, Y_2, and Z_1		---	---	---	---
End points: (Same as subgroup 2)							
Subgroup 4			20				
Terminal strength (lead fatigue)	2036	Test cond. E		---	---	---	---
End points: Seal (leak-rate)	1071	Test cond. G or H for fine leaks; test cond. A, C, D, or F for gross leaks		---	---	$1x10^{-7}$	atm cc/sec
Subgroup 5			15				
Salt atmosphere	1041			---	---	---	---
End points: (Same as subgroup 2)							
Subgroup 6			7				
High-temperature life (nonoperating)	1032	T_{stg} = +100° C; time = 340 hours (see 4.3.4)		---	---	---	---

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TABLE II. Group B inspection - Continued

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 6 - Continued							
End points:							
Collector to base cutoff current	3036	Bias cond. D; V_{CB} = -40 Vdc		I_{CBO}	---	-100	µAdc
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = -12 Vdc; I_E = 1.5 mAdc		h_{fe}	15	220	
Subgroup 7			7				
Steady-state operation life	1027	T_A = +25° C; P_T = 120 mW; V_{CB} = -12 Vdc; time = 340 hours (see 4.3.4)		---	---	---	---
End points: (Same as subgroup 6)							

TABLE III. Group C inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			λ = 10				
High-temperature life (nonoperating)	1031	T_{stg} = +100° C (see 4.3.4)		---	---	---	
End points: (Same as subgroup 6 of group B)							
Subgroup 2			λ = 10				
Steady-state operation life	1026	T_A = +25° C; P_T = 120 mW; V_{CB} = -12 Vdc (see 4.3.4)		---	---	---	---
End points: (Same as subgroup 6 of group B)							

5. PREPARATION FOR DELIVERY

5.1 See MIL-S-19500, section 5.

6. NOTES:

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

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6.2 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - EC
Air Force - 17

Review activities:
Army - MU, MI
Navy - SH
Air Force - 11, 85
DSA - ES

User activities:
Army - SM
Navy - CG, MC, OS, AS
Air Force - 13, 15, 19, 70, 80

Preparing activity:
Army - EL

Agent:
DSA - ES

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-S-19500/27E, SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, GERMANIUM, HIGH-FREQUENCY TYPE 2N384

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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